

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

● ANSWER KEY

Problem-Solving and Data Analysis

12

10 answers · Rationale-rich · Teach from doc

Q1**B****B. 125**

Choice B is correct. According to the graph, when the car is traveling at 50 miles per hour, the braking distance is approximately 225 feet, and when the car is traveling at 30 miles per hour, the braking distance is approximately 100 feet. The difference between these braking distances is $225 - 100$, or 125 feet.

Choice A is incorrect and may result from finding the braking distance for 20 miles per hour, the difference between the given speeds. Choice C is incorrect and may result from subtracting the speed from the braking distance at 50 miles per hour. Choice D is incorrect and may result from finding the difference in the braking distances at 60 and 20 miles per hour.

Q2**A****A. 1 to 4**

Choice A is correct. It's given that Shaquan has 7 red cards and 28 blue cards. Therefore, the ratio of red cards to blue cards that Shaquan has is 7 to 28. This ratio can be reduced by dividing both parts of the ratio by 7, which yields the ratio 1 to 4.

Choice B is incorrect. This is the ratio of blue cards to red cards that Shaquan has. Choice C is incorrect and may result from a calculation error when reducing the ratio. Choice D is incorrect. This may result from finding the ratio of blue cards to red cards, or 28 to 7, and then making a calculation error when reducing the ratio.

Q3**B****B.** 60%

Choice B is correct. According to the table, the minimum wage in 1960 was \$1.00 per hour, and in 1970 it was \$1.60 per hour. The percentage change is therefore $100\left(\frac{1.60 - 1.00}{1.00}\right) = 60\%$.

Choice A is incorrect and may result from averaging the two wages before calculating the percentage change. Choice C is incorrect. This is the 1960 wage expressed as a percentage of the 1970 wage, not the percentage change between the two. Choice D is incorrect and may result from a calculation error.

Q4**B****B.** $\frac{65}{140}$

Choice B is correct. According to the table, 140 students from the two high schools completed summer internships in 2010. Of these, 65 were from Valley High School. Therefore, of the students who completed summer internships in 2010, $\frac{65}{140}$ represents the fraction who were from Valley High School.

Choice A is incorrect. This is the difference between the numbers of students from the two high schools who completed internships in 2010 divided by the total number of students from the two schools who completed internships that year. Choice C is incorrect. This is the fraction of students from Foothill High School who completed internships out of all the students who completed internships in 2010. Choice D is incorrect. This is the number of students from Valley High School who completed internships in 2010 divided by the number of students from Foothill High School who completed internships in 2010.

Q5

B

B. 77

Choice B is correct. The median of a data set with an odd number of data values is defined as the middle value of the ordered list of values. The data set shown has nine values, so the median is the fifth value in the ordered list, which is 77.

Choice A is incorrect. This is the minimum value of the data set, not the median.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the mean of the data set, not the median.

Q6

B

B. Increasing exponential

Choice B is correct. The graph of function f shows that as x increases, $f(x)$ also increases, which means that f is an increasing function. The graph of function f is curved upward and increases at an increasing rate, not a constant rate. This behavior is characteristic of exponential functions, not linear functions. Therefore, of the given choices, function f is best described as increasing exponential.

Choice A is incorrect. For a decreasing function, as x increases, $f(x)$ decreases rather than increases.

Choice C is incorrect. For a decreasing function, as x increases, $f(x)$ decreases rather than increases, and the graph of a linear function isn't curved.

Choice D is incorrect. The graph of a linear function isn't curved.

Q7

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Answer not provided in source data — see rationale below.

Choice C is correct. If the butterfly traveled 2,100 miles in 120 days, then it traveled, on average, $\frac{2,100 \text{ miles}}{120 \text{ days}} = 17.5$ miles per day.

Choice A is incorrect. This is approximately the average amount of time, in days, it took the butterfly to fly one mile: $\frac{120 \text{ days}}{2,100 \text{ miles}} = 0.057$ days per mile. Choice B is incorrect and may result from an arithmetic error. Choice D is incorrect. This is the number of hours in a day rather than the number of miles flown per day.

Q8

B

B. 15

Choice B is correct. It's given that there are 250 trees in a park and of these trees, 6% are birch trees. The number of birch trees in the park can be calculated by multiplying the number of trees in the park by $\frac{6}{100}$. Therefore, the number of birch trees in the park is $250 \cdot \frac{6}{100}$, or 15.

Choice A is incorrect. This is the percentage of trees in the park that are birch trees, not the number of birch trees in the park.

Choice C is incorrect. This is 30%, not 6%, of 250.

Choice D is incorrect and may result from conceptual or calculation errors.

Q9**D**

D. $\frac{107}{135}$

Choice D is correct. If a member of the organization is selected at random, the probability that the selected member is at least 40 years old is equal to the number of members who are at least 40 years old divided by the total number of members. According to the table, there are a total of 135 members of the organization, and 107 of these members are at least 40 years old. Therefore, the probability that the selected member is at least 40 years old is $\frac{107}{135}$.

Choice A is incorrect. This is the probability that the selected member is less than 40 years old.

Choice B is incorrect. This is the probability that the selected member lives east of the river.

Choice C is incorrect. This is the probability that the selected member lives west of the river.

Q10**B**

B. 6

Choice B is correct. The dot plot gives the diameter, to the nearest inch, of each sea star in a group of tide pools. The number of dots above a given diameter represents the number of sea stars with that diameter, to the nearest inch. There are 6 dots above 16 inches. Therefore, 6 sea stars had a diameter, to the nearest inch, of 16 inches.

Choice A is incorrect and may result from calculation or conceptual errors.

Choice C is incorrect. This is the number of sea stars with a diameter, to the nearest inch, of 17 inches, not 16 inches.

Choice D is incorrect. This is the number of sea stars with a diameter, to the nearest inch, of 18 inches, not 16 inches.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

Answers recovered from rationale for Q7 — the source CB data omitted a structured key for these items, so the answer slot shows —. The worked solution in the rationale below each entry contains the correct value.

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